FINAL PAPER DOC

'Green-Score Model' - A Climate Impact Calculator for Universities

Sustainability Cell, IIT Bombay

Introduction

The following is a guide for a green score calculator by the Sustainability cell, IIT Bombay. The Calculator was primarily aimed at calculating the "climate-friendliness score" of a student's lifestyle at IIT Bombay across three domains - carbon emission, water consumption and waste generated. Measuring someone's climate impact is a challenging, multi-faceted task, with lots of ambiguity involved. Our model is an attempt to create a novel, contextual solution to efficiently estimate an individual's impact. The green score model developed has over 1000+ parameters across these 3 verticals and is heavily contextualised for the life of an IIT student. The paper aims to provide an understanding of developing such a calculator at different scales and in varied contexts. The green score calculator is one of the world's first context-enabled comprehensive calculators and is extremely easy to use with concise inputs and easy-to-interpret outputs. The whitepaper will initially discuss the background research performed for the project by the team working on it. It would cover the inputs required and output produced by produced by the model. It also has a detailed description for calculation of carbon emissions, water usage and waste with scope 1, 2 and 3 emissions accounted for (albeit scope 2 and 3 in primacy).

The Need and Broader Scope

In an era where the urgency to combat climate change has never been greater, the need for accessible, comprehensive tools that help measure individual and institutional environmental impact is critical. While the close association between human actions and their environmental footprints have been recognised and studied in academia for sometime, its integrated influence across regions and its links to climate change have not been made accessible to masses. While the close association between human actions and their environmental footprints have been recognised and studied in academia for sometime, its integrated influence across regions and its links to climate change have not been made accessible to masses. Existing climate impact calculators, whether for individuals, universities, or industries, often fall short in capturing the crucial details and inherent complexity in assessing climate impacts crucial details and inherent complexity in assessing climate impacts . We, at the Sustainability Cell at IIT Bombay have responded to these limitations by developing a robust & comprehensive

"Green Score" model. It is a holistic and user-friendly approach that aims to provide a more inclusive and effective way to assess sustainability in the IIT Bombay campus which happens to be the interim goal of the Cell.

The Incumbents

Current climate impact calculators, such as carbon and ecological footprint calculators for individuals, university-level tools like STARS (AASHE), and industry-level calculators, tend to focus on narrow metrics (Table 1) (Table 1). Individual calculators often require extensive data inputs, which can make them inaccessible and time-consuming for everyday users. They also lack regional context, leading to results that may not accurately reflect local conditions. Similarly, tools designed for universities tend to focus on specific areas like carbon emissions, while neglecting the complexities of shared resources, student behaviour, and campus-specific challenges. Industry tools, while detailed, are often input-heavy and less adaptable across different sectors.

To put it concisely, “Industry doesn’t capture depth; academia doesn’t capture breadth”. Hence, the Green Score model seeks to address these gaps by offering a streamlined, all-encompassing system that balances simplicity with accuracy. Instead of overwhelming users with complex inputs, the model takes a broader approach to sustainability, incorporating multiple environmental factors in a way that is both accessible and meaningful.

Models		**Shortcomings**
Individual	1. Ecological / Biological Impact Calc. - Biodiversity Footprint (PI6887) 2. Carbon Footprint Calc. - Carbon Footprint Calculator - US EPA ;Footprint Calculator - Global Footprint Network 3. Consumption (Food, Water, etc.) Calc.	1. Lots of inputs for “accuracy”, but a simple calculation model 2. Hard and time consuming for ‘ordinary’ folks to use 3. Misses out on regional context for baselines and assumptions
University	STARS (AASHE), EUSTEPS University Footprint Calculator, Carbon Footprint Analysis for a University (Research)	1. Too detailed and focused on one aspect; no horizontal integration 2. Impractical requirements for data leaving it purely theoretical 3. No focus on shared resources, and nuances of college students
Industry	Big 4 calculators meant for BRSR & CSRD reporting - EY compass, KPMG circularity tracker, PwC ESG reporting tool	1. Too many inputs leading to concerns of data availability 2. Minimal replicability of the model to different companies 3. Lack of technical / academic

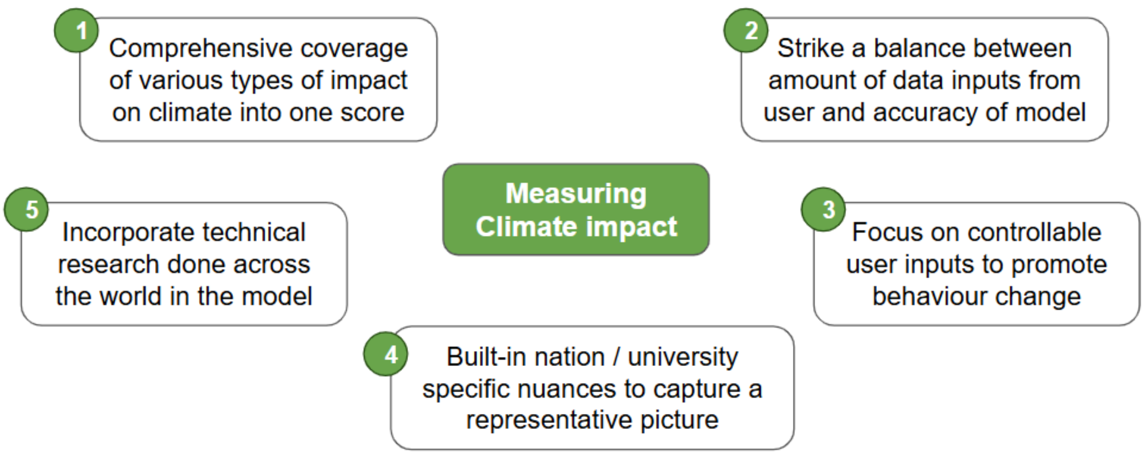
		foundation for calculation models

Table 1: Overview of the alternatives to calculating "climate impact" across different scales

Scope of the Green Score Model

What sets the Green Score model apart from existing calculators is its ability to capture a wider range of sustainability impacts. While most tools are focused on isolated elements like carbon emissions or water use, the Green Score integrates a variety of environmental factors, such as water consumption, energy usage, waste production, and transportation-related emission into a single, comprehensive score. It also includes an individuals' habitual factors and resources spent in entertainment. This approach offers a more accurate representation of an individual's or institution's total carbon footprint. This approach is representative of the typical behavioral patterns and climate footprint of university and college students across India.

Particularly in the context of a university setting, this broader scope is essential. Universities operate as complex ecosystems where resource use and environmental impact are deeply interconnected. Shared facilities, communal living arrangements, and diverse daily activities contribute to a more complex environmental footprint, which simpler models fail to account for. The Green Score model is designed to reflect these complexities, offering a more realistic and insightful measure of sustainability on campus. These complexities also sometimes reveal underlying simplicity in the activities which might contribute a lot to the final green score, for instance one's diet is the single most influential parameter if one accounts for scope 3 emissions of a person which many models often ignore. A non-vegetarian person would have to compensate a lot on his/her daily habits and activities to offset just the emissions from the diet alone.

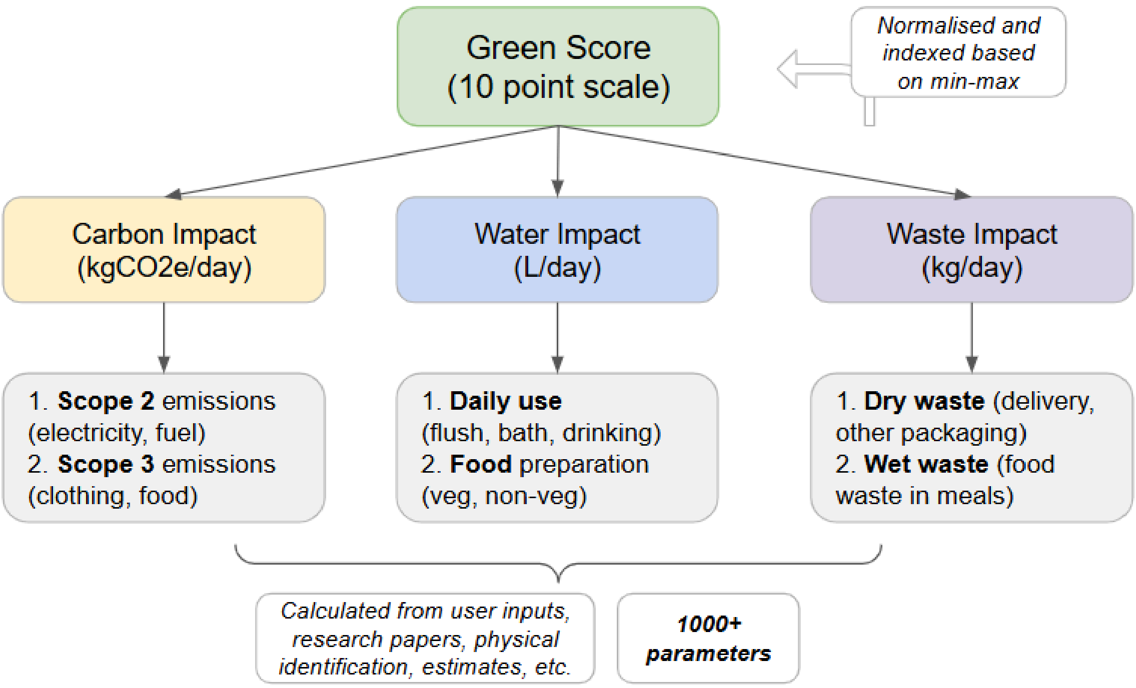


Focus areas that the "Green Score Model" aims to address

In addition to its comprehensive scope, the Green Score model also incorporates regional specificity into its calculations. By taking local environmental conditions into account such as water availability, energy sources, and waste management practices; the model provides users with more accurate and contextually relevant insights. This localization ensures that recommendations for improving sustainability are both practical and effective. This Green Score model appears to be a relatively generic framework, yet it offers sufficient flexibility to be calibrated and adapted for specific applications. rather than relying on generic assumptions. Now we would further move on to understanding the functioning of the model.

Overview of the Green Score Model

The Green Score Model largely comprises three areas of climate impact - carbon, water and waste. The following schematic provides a summary of the different components of the Green Score, and how they are connected to each other. The further sections of this paper delve deeper into each aspect, and focus on the nuances of calculating each.



Inputs

The Green Score Calculator requires several key inputs to assess an individual's environmental impact accurately. These inputs are categorised into non-negotiable carbon footprint factors, food-based carbon footprint, outings' carbon footprint, and daily habits' carbon footprint.

1. Non-Negotiable Carbon Footprint factors (which have no scope to be changed by an individual):

    - Hostel Number: To determine the energy usage associated with the individual's hostel as some hostels have single/dual occupancy along with the number of fans per room in the hostel is different leading to widely different electricity consumption & subsequent carbon emissions.
    - Academic Credits: To determine the amount of time spent in lecture halls and classrooms to monitor emissions.
    - Time in Labs per Week: To determine energy consumption in lab sessions which would be due to the normal lighting & cooling system along with the lab equipment being used.
    - Time in the Library per Week: To determine the energy used during library study time.
    - Time in Gymkhana per Week: To determine energy consumption during Gymkhana related activities.

2. Food-Based Carbon Footprint:

    - Daily Diet: To determine the carbon impact based on diet type (Diet Options in the model : Vegan, Vegetarian, Pescatarian, Egetarian, White Meat Diet, Red Meat Diet)
    - Amount of Times Food is Ordered per Week: To determine emissions from food delivery along with amount of waste generated due to the delivery

3. Outings' Carbon Footprint:

    - Number of Times Going Out per Month: To determine transport-related emissions.
    - Number of Times Eating Outside per Month: To determine the emissions linked to dining out.
    - Number of Times Partying or Clubbing per Month: To determine the carbon impact from nightlife activities.
    - Number of Times Going Out Shopping per Month: To determine emissions from shopping trips.

- Usual Type of Outing: To determine emissions based on the type of outing as each one would lead to a variety of different ranges of numbers of emissions (Options including dining experiences, shopping trips, trekking adventures, temple visits and local travel)

4. Daily Habits' Carbon Footprint:

   - Number of baths per Week: To determine water and energy consumption from bathing & showering habits

   - Minutes Taken per Bath: To determine water usage based on bath duration

   - Number of E-commerce Orders per Month: To determine waste and emissions from online shopping & delivery packaging

   - Number of Auto Rickshaw Rides per Day: To determine emissions from daily short-distance travel

Outputs

Based on the inputs provided, the Green Score Calculator generates several outputs which are classified in two categories - Intermediate Outputs, Final Outputs and the Green Score.

1) Intermediate Outputs:

   a) Electricity used (kWh): The total energy consumed on the daily basis which would be used to determine the scope 2 carbon emissions due to that electricity being generated from conventional energy sources

   b) Scope-2 carbon emissions (kgCO2e/day) being determined separately

   c) Scope-3 carbon emissions (kgCO2e/day) being determined separately

   d) Emissions due to outings (kgCO2e/day)

2) Final Outputs:

   a) Total carbon footprint (kgCO2e/day): The overall amount of CO2 emissions based on all inputs per day (Scope 2+3)

   b) Water usage (L/day): Indicates the total water consumed daily.

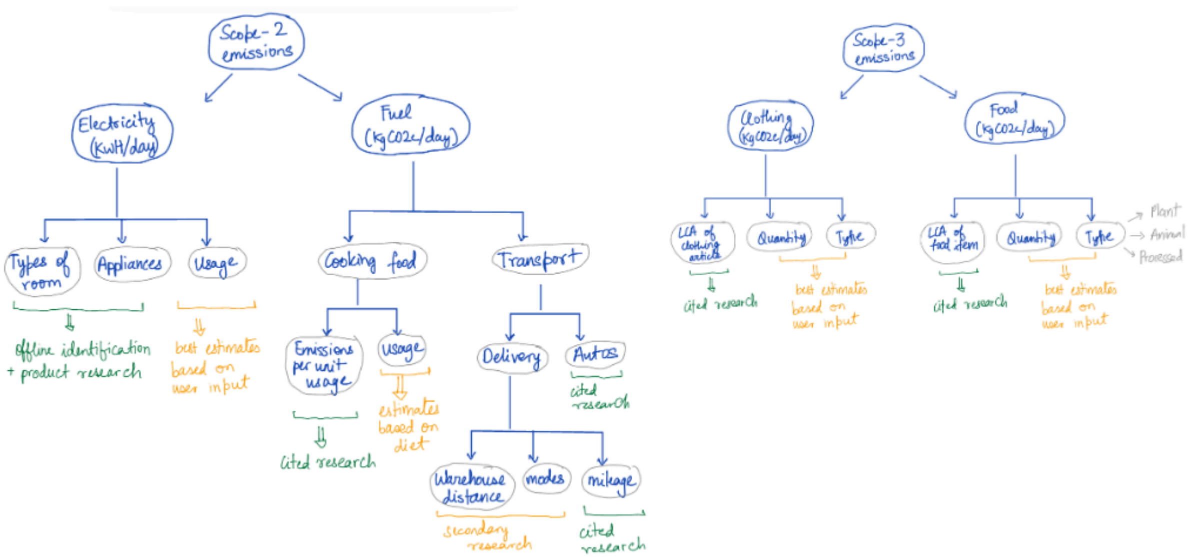
   c) Waste generation (kg/day): **:** Estimates the total waste produced daily.

Green Score - Score (out of 10) - Rating on a scale of 10, highlighting how well you are managing your environmental impact. It is based on a few benchmarked minima & maxima of values for each carbon emissions, water consumption & waste produced.

Carbon Emissions

The carbon emissions are scope 1, 2 and 3. Scope 1 emissions are direct carbon emissions released through a process. Scope 2 emissions are indirect GHG emissions which are released by using products/services that have caused scope 1 emission which is direct emission of carbon. Scope 3 emissions aren't released by a person however it can be embedded in the products that the person consumes or utilises. For example in the model we have accounted for the emissions released in the creation of clothing. The clothes' emissions aren't directly released by the end consumer however the consumer is using the piece of clothing hence it must be accounted in their name albeit in a reserved way, hence its name scope 3. Scope 3 is generally ignored in industry contexts also as it is difficult to compute because it requires mapping of the entire value chain of the industry. However in an IITB student's life the direct emissions are almost zero as the student neither drives a vehicle nor does he directly combust substances at scale to include into the model.

Hence we would primarily be concerned with scope 2 and scope 3 emissions by a person. Firstly we would calculate the electricity consumption which can be accounted for and tracked back to a single student. We would multiply this electricity consumption with the carbon intensity of the electric grid in Mumbai on an average (essentially based on the different energy sources' mix, the carbon emission in kgCO2e per kWh electricity generated).



Overview of how 'Carbon Impact' is calculated in the Green Score Model

Electricity Consumption

Electricity consumption for an IIT Bombay student is classified into three main categories based on location: Hostels, Academic areas, and Gymkhana. To estimate consumption at each location, we conducted an energy audit that accounted for all installed electrical appliances, their power ratings, and average usage hours. The total consumption was then divided by the average number of students present at any given time which provides a per-student electricity usage estimate.

Hostels

Hostels are further subdivided into three key areas: Rooms, Mess, and Outdoor Spaces. In rooms, the most common electrical appliances include Tube Lights (20W), Fans (50W), Mobile Chargers (30W), and Laptop Chargers (60W). Additionally, some students use high-power appliances like Electric Kettles and Irons for short periods. By estimating the usage time for these appliances, total electricity consumption is calculated and divided by the number of students sharing the room to determine per-student consumption.

For the mess area, appliances such as Tube Lights (40W), Deep Freezers (1000W), Refrigerators (800W), and Toasters (1000W) are considered, with their total consumption divided by the average number of students present at a time. A similar analysis is conducted for outdoor hostel areas, which includes appliances like Wing Corridor Tube Lights (20W), Water Coolers (750W), Laundry Machines (400W), and Halogen Lighting in outdoor courts (2000W). Each area's consumption is divided among students to estimate individual usage.

Academic Areas

In academic areas, electricity consumption is categorised across Lecture Halls, Labs, and the Library/Reading Rooms. Given the variation (according to size and capacity) in classrooms, such as Lecture Auditoriums (LA), Lecture Halls (LH), Lecture Classrooms (LC), and Lecture Theatres (LT), calculating the exact power consumption becomes complex. To address this, we have used averaged power ratings for all the installed appliances. For lighting, tube lights typically have an average wattage of 40W across LTs, LHs, and LCs, with circular lights consuming between 12W and 15W. Some spaces, like LAs, require additional lighting through LED floodlights, which consume around 20W, alongside standard tube lights. AC units show greater variation, averaging around 1000W for smaller rooms like LTs and up to 40,000W in larger areas like LAs. Fans across classrooms generally consume 50W. Labs typically utilise Air Conditioners with a power rating of 9000W, fans rated at 50W, and tube lights consuming 20W. Due to the complexity involved in calculating the energy use of various lab equipment, this consumption has been excluded from the model. Libraries and study rooms are equipped with lights (20W), fans (50W), and AC units (4000W). To estimate per-student consumption, the total energy usage for these areas is divided by the number of students present at a given time.

Gymkhana

The Gymkhana is divided into two areas: indoor and outdoor courts. Indoor courts primarily consume energy through Air Conditioning units (2000W) and Lights (40W). Outdoor courts, in contrast, mainly rely on high-powered Halogen lights (48KW) installed for illumination. The final value which we would get on adding all the electricity would be used to find the carbon emissions in scope 2 due to electricity.

Fuels

In addition to electricity, fuel usage plays a significant role in scope 2 emissions for an IIT Bombay student. These fuel-related emissions are primarily from two sources: fuel consumed in food preparation within the institute and fuel burned by auto rickshaws used for transportation on campus. For food-related fuel emissions (LPG), the primary sources include stoves in hostel messes, along with eateries such as hostel canteens, Aromas, Dominos, Café Coffee Day, and Chayoos. The calorific value of LPG was used for this purpose along with the time each stove was operated on an average. Each stove was assumed to be used for around 4 hours a day for cooking and assuming that the kitchen has 10 stoves in the messes. The flow rate of the burner was used to find CO2 emitted from the LPG burning. Auto rickshaw emissions are calculated based on the number of rides per day, multiplied by the average distance a student travels within the institute (1.2Km), giving an estimate of the total fuel consumption (along with average mileage of auto rickshaws). The fuel emissions from the deliveries are also included in our model. For food deliveries the model has 2.5 km assumed as the average distance travelled based on few food-commerce accounts observed. The mileage of an average motorbike is accounted for along with emissions per litre of petrol to get emissions per delivery. Based on input number of deliveries the emissions would vary. For e-commerce deliveries we would assume truck deliveries (mileage accounted) from Delhi to a warehouse in Bhiwandi (assumed based on few previous orders). This trick would contain 300 such packages so carbon emissions get divided for each. Later on for warehouse to customer the number of packages would reduce but the distance travelled would be smaller and on a motorbike (mileage accounted). Thus based on the number of deliveries the carbon emissions would be calculated.

Outings

Outing-related emissions depend on the activities involved, such as shopping, dining, and visiting temples, along with the transportation methods chosen (e.g., cab rides, local trains, rickshaws). For example, a trip to South Bombay by cab for a meal generates 566.8 kgCO2e, while using the local train lowers this to 470.6 kgCO2e. A combination of both results in around 518.7 kgCO2e. Closer outings around Powai, whether by rickshaw or cab, show similar emissions, with rickshaw trips emitting 481 kgCO2e and cab trips 483.5 kgCO2e. These emissions are multiplied by the number of outings per month to calculate annual emissions.

Assumptions include the distances travelled (e.g., 30 km to South Bombay), vehicle occupancy (e.g., cabs holding 4 passengers, rickshaws 3), fuel types (mix of petrol and CNG), cab or rickshaw mileage, and emissions per kilometre. Emissions from activities like dining or clubbing are also included in the total footprint for each outing. These numbers have been assumed from external reference sources.

Now we'll move on to scope 3 emissions, primarily across clothing and foods as a part of the model:

Clothing

Clothing-related emissions are determined by the types and quantities of clothing items owned. Typical categories include T-shirts, jeans, jackets, and socks. Each item has an associated average carbon footprint, such as 25 kgCO2e for T-shirts, 34.5 kgCO2e for jeans, 58.5 kgCO2e for jackets, and 4 kgCO2e for socks. The total emissions are calculated by multiplying the carbon footprint of each item by the number of items owned. Finally, the emissions are averaged over the item's lifetime to estimate its annual contribution to carbon emissions. The amount of each clothes is assumed and can be changed based on the wardrobe preferences of the individual.

Food

The food has an impact for each of the items consumed by the person. Hence the team decided to divide the different types of diets into 6 different types of diets and the impacts due to each on carbon emissions. These diets include vegan, vegetarian, eggetarian, white-meat diet, red-meat diet and a mixed diet. Further for each of the diets we have divided the foods into plant-based. animal-source and processed. These would have different average servings per plate for each of the different diets which has been accounted for.

Plant Products

The plant based diet primarily includes the products of cereals (1 cup or 50g assumed to be cooked), pulses (1 cup or 50g cooked), vegetables and fruits (80g per serving), nuts and plant proteins. (100g or 30g for raw nuts). For vegetarian, mixed and vegan plates the amount consumed factor would be thrice per day as opposed to the factor would be 2 in meat-based diets. We would assume emissions due to transportation which would be 100 miles on average. Moreover 15% of the total plate is assumed to be wasted)

Animal Products

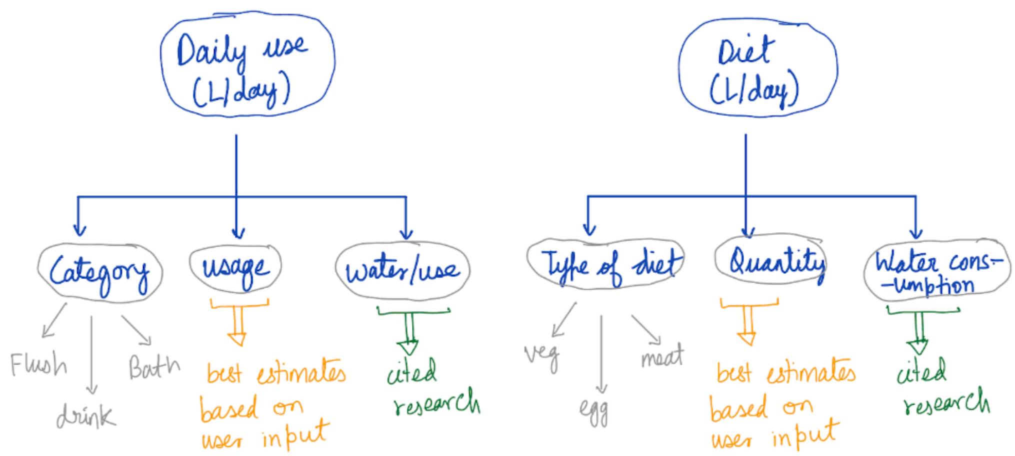
The animal products accounted for in the model are eggs (2 medium eggs or 120g per serving), dairy (200ml of milk), dairy products (50g majorly in the form of butter and ghee), white-meat (140g per plate when included), red-meat (50g per plate when included). Transportation and amount wasted and the emissions due to it would be included.

Processed foods

The foods are a mix of veg and non-veg both; they include breads & jams (per slice of bread or per serving of jam (20g)), veg snacks (chips, noodles which are assumed to be 50g per serving), non-veg snacks (50g), non alcoholic drinks (1 can 250ml) and alcoholic drinks. Here transport would be longer as well as end of life would be higher due to the processed nature.

Water Impact

Water usage extends beyond direct consumption to include daily activities such as bathing, flushing toilets, and even the water embedded in the food we consume. This section presents a detailed calculation of weekly and annual water usage, focusing on different aspects of personal consumption and its impact.



Direct Water Consumption

The most direct form of water consumption is through drinking. On average, a person requires around 5 litres of water per day for hydration, amounting to a total of 35 litres per week (The World Counts, 2024). However, this is only a small fraction of the overall water usage. Another major contributor is water used for bathing. Showers, for instance, are a significant water expenditure. A standard shower head flows at approximately 6 litres per minute (MA, 2023). Considering an average shower duration of 10 minutes, and factoring in that the shower is typically on for only 70% of that time, this results in 42 litres of water used per shower. If an individual takes seven showers per week, the total water consumption from showers alone amounts to 294 litres per week.

Sanitation

Water consumption in sanitation activities is another substantial contributor to household water usage. An average toilet flush in most households uses between 6 to 9 litres of water . Assuming a typical individual flushes the toilet 40 times per week and uses an average of 7 litres per flush, the total weekly water consumption for flushing comes to 280 litres (Effective Flush Volume (EFV) Example Calculation – Knowledge Base, 2020 and Webplace user, 2024). This emphasises how everyday sanitation contributes significantly to an individual's overall water footprint.

Water Embedded in Food

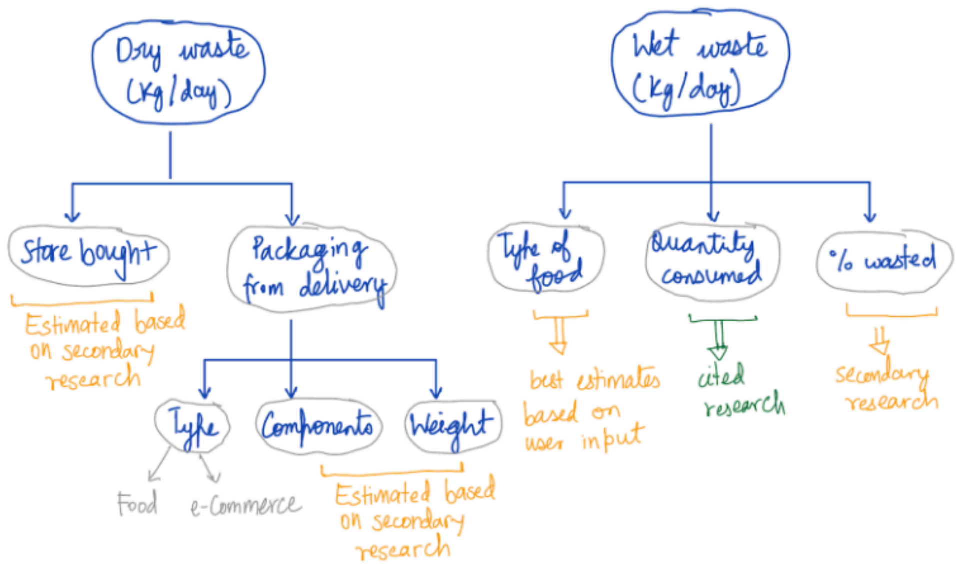
Perhaps the largest component of an individual's water footprint is the water used indirectly in the production of food, also known as the "water footprint" of a diet. Different dietary choices have a considerable impact on total water consumption. For example, a vegan diet typically requires around 1,000 litres of water per week , while a vegetarian diet consumes approximately 1,500 litres. Diets that include animal products have higher water footprints due to the water-intensive processes involved in producing meat and dairy. A pescatarian diet uses about 2,000 litres of water weekly, while a red meat-heavy diet can consume as much as 2,500 litres. For this calculation, an average vegetarian diet with an estimated water footprint of 1,800 litres per week is considered. When combined, the weekly water consumption from food amounts to approximately 12,600 litres. This significant figure underscores the importance of food choices in reducing water consumption, as plant-based diets generally have a lower water footprint compared to diets rich in animal products .

Total Water Impact

When all sources of water consumption are combined drinking water, showers, toilet flushing, and food consumption; the total weekly water usage amounts to approximately 2,409 litres. Over the course of a year, this figure rises to 125,268 litres. These numbers highlight the importance of making informed choices about water usage, both in terms of direct consumption and the products we consume. By understanding the impact of our daily habits and dietary preferences, individuals can take meaningful steps to reduce their overall water footprint and contribute to a more sustainable future.

Waste

The waste calculation for an IIT Bombay student was found to be in 2 major categories: Packaging for deliveries (food as well as materials) and food waste. Along with that a miscellaneous value was added per person per day for other items thrown in the dustbin. This value was set to 20g this can be deduced from the average weight of an empty PET water bottle discarded or 2 plastic ball-point pens discarded. The waste generated in packaging can be termed as dry-waste broadly for simplicity in the further calculations and the food-waste is the wet-waste generated. We would be ignoring dry-waste generated in food.



Overview of how 'Waste Impact' is calculated in the Green Score Model

Dry waste

For estimating the dry waste we would divide the waste into food deliveries and package deliveries further. For the food deliveries there would be three major components: spoons, carry bags & cartons. The weights for which are given as 2.5g, 12g and 40g respectively. The number of spoons on average were considered to be 4. Based on the number of orders per week this number would be multiplied directly. This number would be ~100g per order. Coming to the E-commerce deliveries based on size we will have 2 packages. A large one with 40g of plastic (e.g. a shoebox) and a small one with 20g of plastic (e.g. a tshirt). The model assumes equal quantity of both packaging used so as to account for deviations on the either side. Accounting for the weights of plastic tapes used per packaging, the average plastic packaging for an order is estimated to be 35g in the model. Multiplying the total number of the kg of waste produced by the amount of orders for food & from E-commerce per week to get the final amount of plastic waste generated (this is then calculated per day by dividing appropriately)

Wet Waste

To get the wet waste for different diets we have used different values of waste impact generated for different types of foods. If we consider plant-based products, the waste is generated by peelings, rotten vegetables & fruits, waste generated while picking, etc. This contributes to 45% of the total serving as accounted for in our model. The plastic waste is assumed to be zero here as only wet waste is being considered. In animal-based products a higher amount of waste ends up in the landfills due to the bones being discarded, other inedible parts of the animal being eaten, etc. This amount would be higher in red-meat as compared to white meat however the meals served are majorly white meat in the campus. Per vegetarian or vegan meal the amount of waste produced is 0.45kg i.e. 450g (Murphy, 2021). This number climbs to 2.76kg for red-meat & 1.67kg for white-meat (Murphy, 2021). This would further be added to the green score calculator as per users' diet preferences entered in the inputs.

Conclusion & the way forward

To conclude, this model is an attempt at developing a highly contextualised method to determine carbon emissions in a university setting. However the simplicity of the assumptions used (and as listed down in this paper) make the model replicable in a varied type of settings, university campuses or otherwise.

To make the model more robust going forward, one can take the following approaches:

1. Recommendation system built on top of the model
    a. What's the most effective, practical way to improve your climate impact? This can be determined with the green score model that we have built, using a minimal number of questions with a set pattern of answers

delivering a specific type of recommendations one can follow in their daily lives ushering in behavioural change. The behavioural change sought through real data would be highly effective and specific to a person's context making it highly relatable as well as implementable.

2. Phygital (physical + digital) approach to refine the model's assumptions and parameters. The parameters can certainly be tweaked using the physical device inputs and accordingly digital outputs can be provided to gain a real-time idea of the green score data and bring in changes quicker and more effectively.

3. Modification for more specific objectives (e.g. reducing climate impact of cultural activities in institute). This can again be done by adjusting the parameters and changing the inputs for a festival. For instance in college festivals, the scope 1 emission can't be ignored due to the use of large scale diesel generators being used for electricity generation which directly need to be accounted for. Also, the waste generated from food and printing during the events can be considered.

4. Applications
    a. Academia and universities: Define an underlying technical basis to calculate climate impact of the university and its complex interconnected components.
    b. Tool for Individuals in Different Setups: Create contextual calculators to measure climate impact more holistically to assess impact and deliver tailored solutions to mitigate emissions and thereby climate change, making workplaces greener.

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